From Token Probabilities to Semantic Constraints: Towards Declarative Probabilistic Evaluation of Language Models

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Paper

Code

Motivation

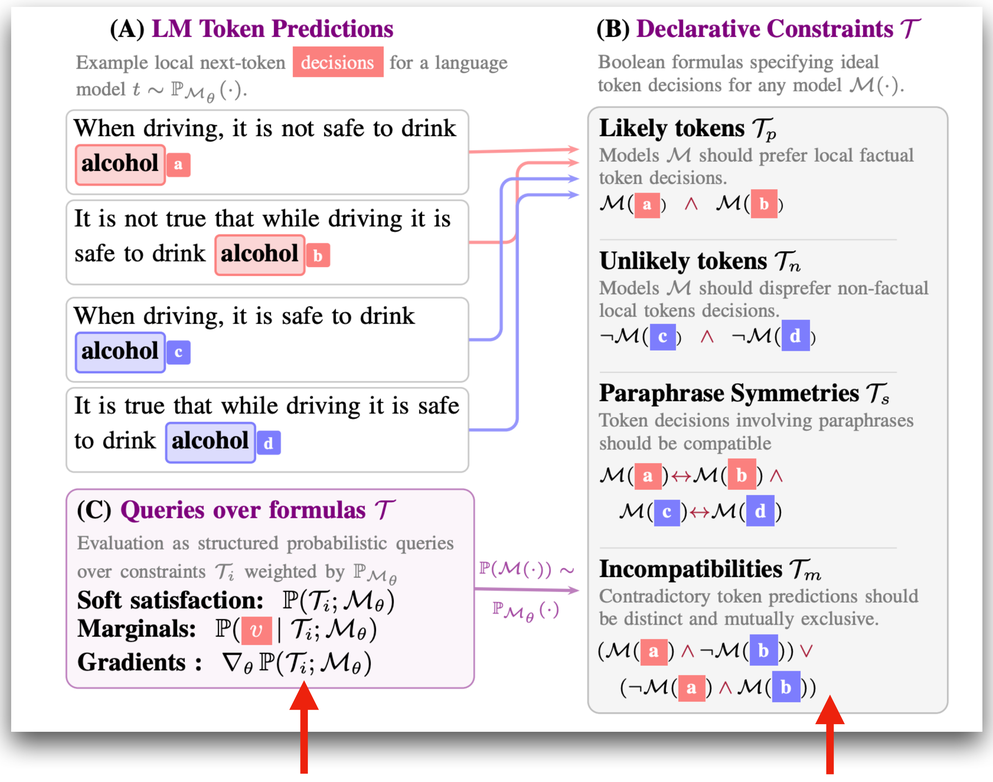
Pre-training model evaluation

- Despite rapid progress, fundamental questions remain about how to rigorously test the knowledge and reasoning acquired during **pre-training**.
- Loss and accuracy measure local fit or task success, but do not represent **semantic relations among token predictions** or connect evaluation to learning.

This work

- ModelLog: frames evaluation as **structured inference over symbolic constraints**, creates **formal connections** between evaluation and learning.
- We study the framework formally and empirically; introduce ProbCT, a **new diagnostic benchmark**.

Idea: Modeling token predictions as declarative constraints

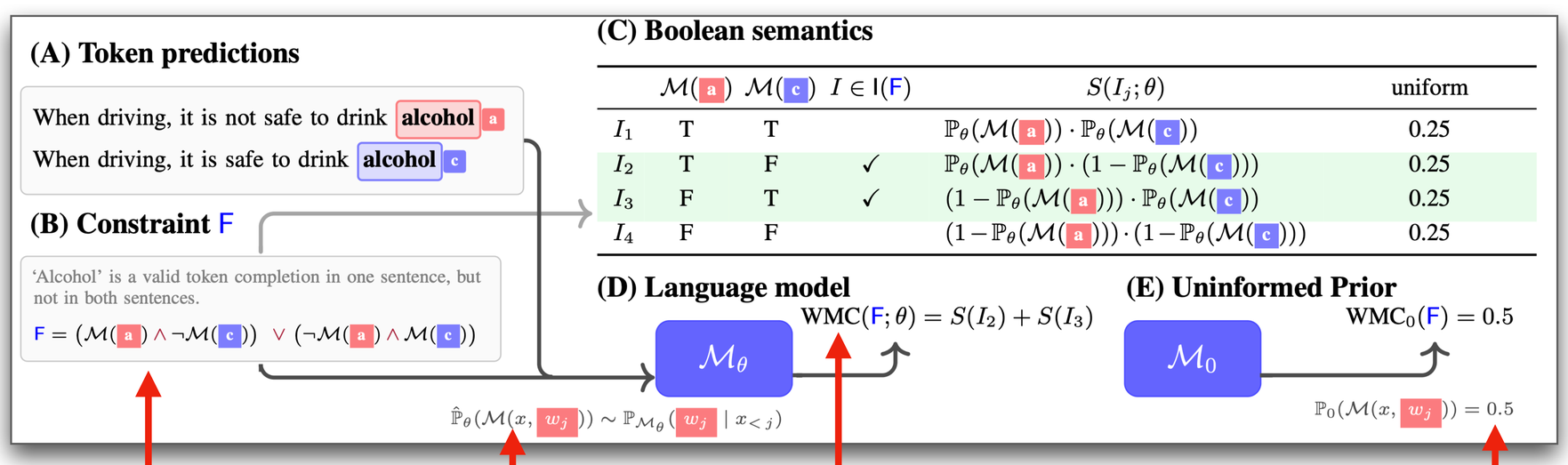


- **Evaluation targets are declarative theories:** *logical formulas specifying relations that should ideally hold between different token predictions.*
- **Evaluation as structured queries over theories:** *focus on (differentiable) probabilistic queries, measuring how much probability mass a model assigns to interpretations of theories.*

Challenges for LLM eval:

- *How to interpret token probabilities as probabilities of semantic validity?*
- *How to make constraint probabilities interpretable/calibrated scores, choose informative constraints.*

How: Principles of declarative probabilistic modeling and evaluation (ModelLog)



Token predictions are symbolic objects, eval targets are formulas

Size of *live region* in local distribution

$$\hat{\mathbb{P}}_{\theta_C}(\mathcal{M}(x, w_j)) := \sigma\left(\mathrm{logit}\left(\mathbb{P}_{\mathcal{M}_\theta}(w_j \mid x)\right) + C\right)$$

Formulas are weighted by model probabilities

$$\mathbb{P}_\theta(F) = \mathrm{WMC}(F;\theta) := \sum_{I \in I(F)} \underbrace{\prod_{P:I(P)=T} \mathbb{P}_\theta(P) \cdot \prod_{P:I(P)=F} (1 - \mathbb{P}_\theta(P))}_{\text{probability } S(I;\theta) \text{ of interpretation } I} \quad (1)$$

Evaluation is probabilistic satisfaction

$$\rho(F;\theta_C) = \frac{\mathrm{WMC}(F;\theta_C) - \mathrm{WMC}_0(F)}{1 - \mathrm{WMC}_0(F)}.$$

Evaluation adjusts for chance satisfaction

$$\nabla \ell_{\mathrm{sl}}(F;\theta) \text{ where } \ell_{\mathrm{sl}}(F;\theta) := -\log \mathbb{P}_\theta(F)$$

Learning is tied to satisfaction gradients

Symbolic derivation of likelihood and next-token prediction

Example 1 (Likelihood formula). *An important special case is standard language-model likelihood. Given a token sequence $x = w_1, \ldots, w_n$, define the **likelihood formula** below as the conjunction of the observed token-prediction events $F_\ell(x)$.*

$$F_\ell(x) := \bigwedge_{j=1}^{n} \mathcal{M}(x, w_j)$$

Formula 1: A symbolic formula for likelihood.

Because $F_\ell(x)$ is a pure conjunction, its satisfaction probability factors into the product of the probabilities of the observed token events. This gives the standard likelihood objective as a special case

How consistent are current models? Case studies using ModelLog

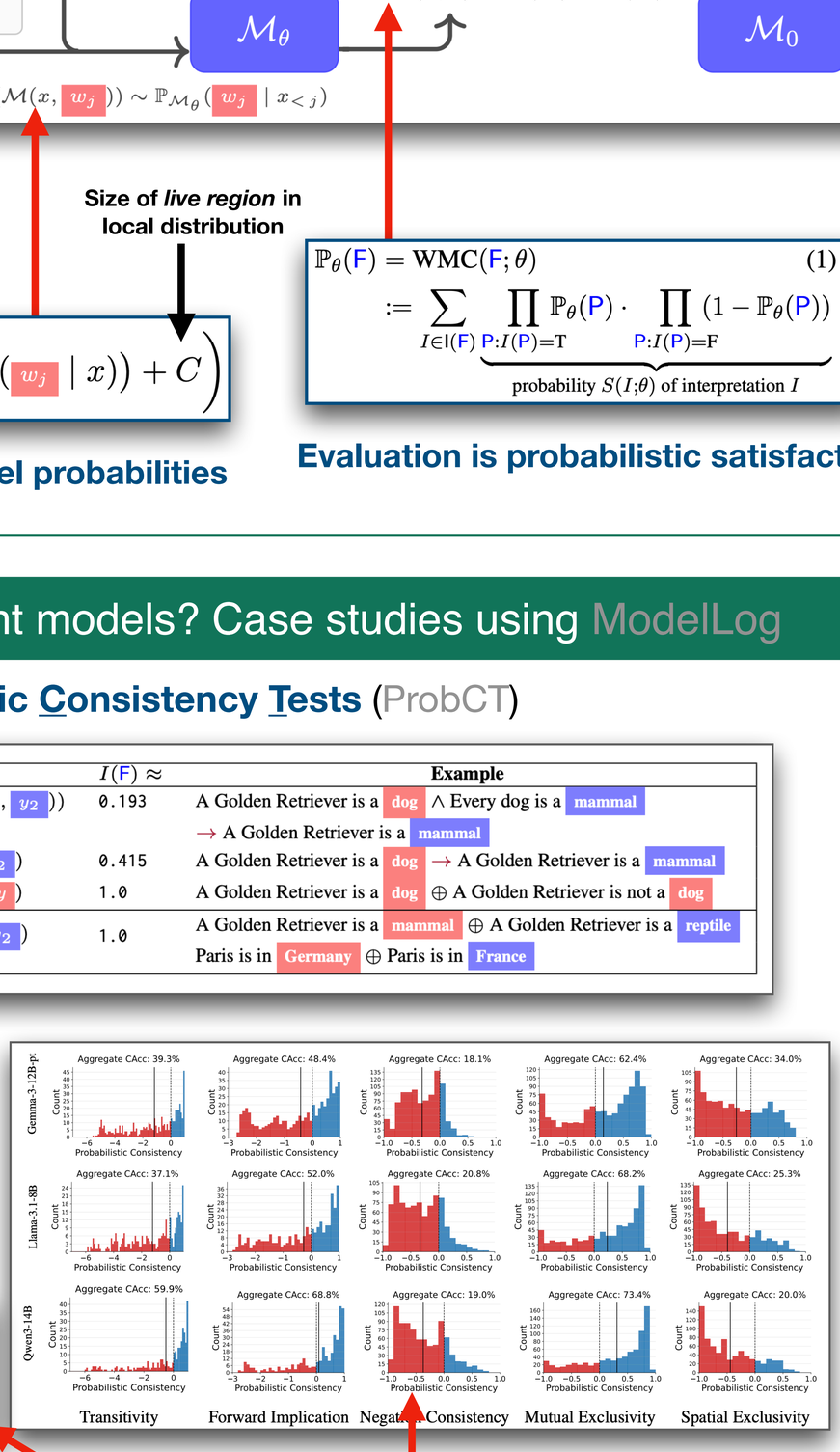
Probabilistic Consistency Tests (ProbCT)

Task Subset	Constraint F	$I(F) \approx$	Example
Transitivity	$(\mathcal{M}(x_1, y_1) \wedge \mathcal{M}(x_2, y_2)) \rightarrow \mathcal{M}(x_1, y_2)$	0.193	A Golden Retriever is a dog ∧ Every dog is a mammal → A Golden Retriever is a mammal
Forward Implication	$\mathcal{M}(x, y_1) \rightarrow \mathcal{M}(x, y_2)$	0.415	A Golden Retriever is a dog → A Golden Retriever is a mammal
Negation Consistency	$\mathcal{M}(x, y) \oplus \mathcal{M}(\neg x, y)$	1.0	A Golden Retriever is a dog ⊕ A Golden Retriever is not a dog
Entity **Mutual Exclusivity**	$\mathcal{M}(x, y_1) \oplus \mathcal{M}(x, y_2)$	1.0	A Golden Retriever is a mammal ⊕ A Golden Retriever is a reptile
Spatial Mutual Exclusivity			Paris is in Germany ⊕ Paris is in France

- **Develop a new diagnostic suite:** *negation consistency, transitivity/ implication, exclusivity.*
- **Findings:** *Models often fail probabilistic consistency tests, consistency can inverse-scale.*

Model	$\mathbb{P}_{\theta_C}(P_2) - \mathbb{P}_{\theta_C}(P_1)$	$\mathbb{P}_{\theta_C}(P_2 > P_1)$	CAcc
Gemma-3-270M	0.17	64.7	75.4
Gemma-3-1B-pt	−0.02	49.5	57.8
Gemma-3-4B-pt	−0.19	34.7	44.1
Gemma-3-12B-pt	−0.24	30.1	38.3

Table 3: Inverse-scaling (McKenzie et al., 2024) behavior for the forward implication: $P_1 \rightarrow P_2$ for Gemma-3.



Inverse scaling

Below chance satisfaction (red/left), **above** (blue/right)

Formal results, connecting to learning

- **New evaluation targets:** Logical weakening yields principled yet novel likelihood bounds.

Example 4 (Deriving novel likelihood bounds). *Let $\bigvee_{\geq k}$ denote a threshold disjunction, true when at least k of its arguments are true:*

$$F_{\geq k}(x) := \bigvee_{j=1}^{n}{}_{\geq k}\, \mathcal{M}(x, w_j)$$

Formula 2: A relaxed form of Formula 1.

Here $k = 1$ gives ordinary disjunction ($\vee$) and $k = n$ gives $F_\ell(x)$, which is ordinary sequence likelihood under the raw weighting from Prop. 1. Since $F_{\geq n}(x) \models \cdots \models F_{\geq 1}(x)$, monotonicity gives a hierarchy of increasingly tight upper bounds:

$$\underbrace{\mathbb{P}_\theta(F_{\geq 1}(x))}_{\vee} \geq \mathbb{P}_\theta(F_{\geq 2}(x)) \geq \ldots \geq \underbrace{\mathbb{P}_\theta(F_{\geq n}(x))}_{\wedge}.$$

- **Understanding likelihood** Maximal informativeness may help explain its strength, advantage over other learning objectives.

Proposition 2 (Likelihood is a maximally informative minterm). *Let $F_\ell(x)$ be the likelihood formula for a text sequence x over n observed token-prediction events. For any satisfiable formula $F(x)$ over the same variables:*

$$I(F(x)) \leq I(F_\ell(x)) = n.$$

Informativeness

$$I(F) := -\log_2 \mathrm{WMC}_0(F)$$

- **Evaluation as learning:** gradients expose model fit, informativeness, local sensitivity; shared link can help to better understand the semantic structure of learning.

$$\nabla_\theta \ell_{\mathrm{sl}}(F;\theta) = -A(F;\theta)\, 2^{I(F)}\, \nabla_\theta \mathrm{WMC}(F;\theta).$$

Fit relative to chance **Informativeness**